UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

*2934833602*

**MATHEMATICS (SYLLABUS D)** **4024/01**

Paper 1 **May/June 2008**

**2 hours**

Candidates answer on the Question Paper.

Additional Materials: Geometrical instruments

**READ THESE INSTRUCTIONS FIRST**

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

If working is needed for any question it must be shown in the space below that question.
Omission of essential working will result in loss of marks.

**NEITHER ELECTRONIC CALCULATORS NOR MATHEMATICAL TABLES MAY BE USED IN THIS PAPER.**

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.
The total of the marks for this paper is 80.

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**NEITHER ELECTRONIC CALCULATORS NOR MATHEMATICAL TABLES MAY BE USED IN THIS PAPER.**

**1** Evaluate

**(a)** $\frac{1}{2} - \frac{3}{7}$,

**(b)** $2\frac{2}{3} \times 1\frac{3}{4}$.

*Answer* (*a*) .....[1]

(*b*) .....[1]

---

**2** Evaluate

**(a)** 25 – 18.3,

**(b)** 1.7 × 0.03.

*Answer* (*a*) .....[1]

(*b*) .....[1]

---

**3** It is given that $f(x) = 5x + 2$.

Find

**(a)** $f(-2)$,

**(b)** $f^{-1}(x)$.

*Answer* (*a*) .....[1]

(*b*) $f^{-1}(x)$ = .....[1]

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**4** By writing each number correct to 1 significant figure, estimate the value of

$$\frac{8.62 \times 2.04^2}{0.285}.$$

*Answer* .....[2]

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**5** It is given that $68.2 \times 0.235 = 16.027$.

Hence evaluate

**(a)** $0.0682 \times 2350$,

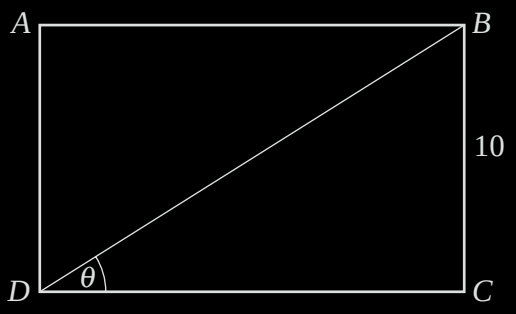
**(b)** $160.27 \div 0.0235$.

*Answer* (*a*) .....[1]

(*b*) .....[1]

---

**6**




| | |
|---|---|
| $\sin\theta$ | $\frac{5}{13}$ |
| $\cos\theta$ | $\frac{12}{13}$ |
| $\tan\theta$ | $\frac{5}{12}$ |

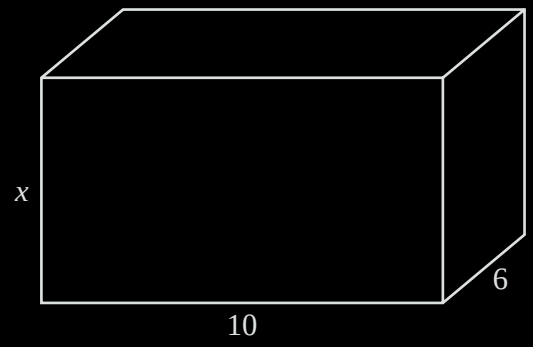
*ABCD* is a rectangle with *BC* = 10 cm.

Using as much information from the table as is necessary, calculate *BD*.

*Answer* *BD* = ..... cm [2]

---

**7** The diagram shows a solid cuboid with base 10 cm by 6 cm.
The height of the cuboid is $x$ centimetres.




**(a)** Find an expression, in terms of $x$, for the total surface area of the cuboid.

**(b)** The total surface area of the cuboid is 376 $cm^2$.

Form an equation in $x$ and solve it to find the height of the cuboid.

*Answer* (*a*) ..... $cm^2$ [1]

(*b*) ..... cm [2]

---

**8** Evaluate

**(a)** $9^0$,

**(b)** $9^{-2}$,

**(c)** $9^{\frac{3}{2}}$.

*Answer* (*a*) .....[1]

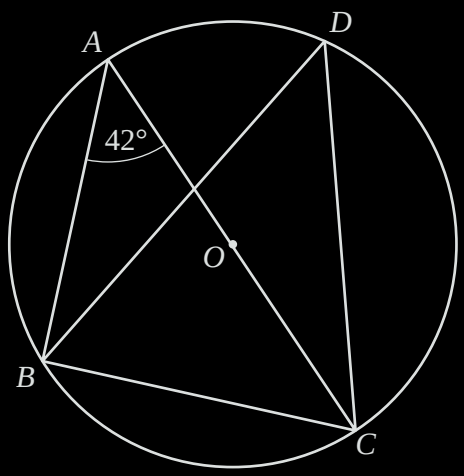
(*b*) .....[1]

(*c*) .....[1]

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**9**




The diagram shows a circle, centre $O$, passing through $A$, $B$, $C$ and $D$.
$AOC$ is a straight line and $B\hat{A}C = 42°$.

Find

**(a)** $B\hat{D}C$,

**(b)** $A\hat{B}C$,

**(c)** $A\hat{C}B$.

*Answer* (*a*) $B\hat{D}C$ = .....[1]

(*b*) $A\hat{B}C$ = .....[1]

(*c*) $A\hat{C}B$ = .....[1]

---

**10** It is given that $y$ is directly proportional to the square of $x$ and that $y = 1$ when $x = \frac{1}{2}$.

Find

**(a)** the formula for $y$ in terms of $x$,

**(b)** the values of $x$ when $y = 9$.

*Answer* (*a*) $y$ = .....[2]

(*b*) $x$ = ..... .....[1]

---

**11** The following list gives the names of six shapes.

| | | |
|---|---|---|
| Square | Rectangle | Equilateral triangle |
| Kite | Trapezium | Parallelogram |

From this list, write down the name of the shape which always has

**(a)** rotational symmetry of order 3,

**(b)** rotational symmetry of order 2 and exactly 2 lines of symmetry,

**(c)** one line of symmetry only.

*Answer* (*a*) .....[1]

(*b*) .....[1]

(*c*) .....[1]

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**12** Solve the simultaneous equations

$$2x - 3y = 13,$$
$$3x + y = 3.$$

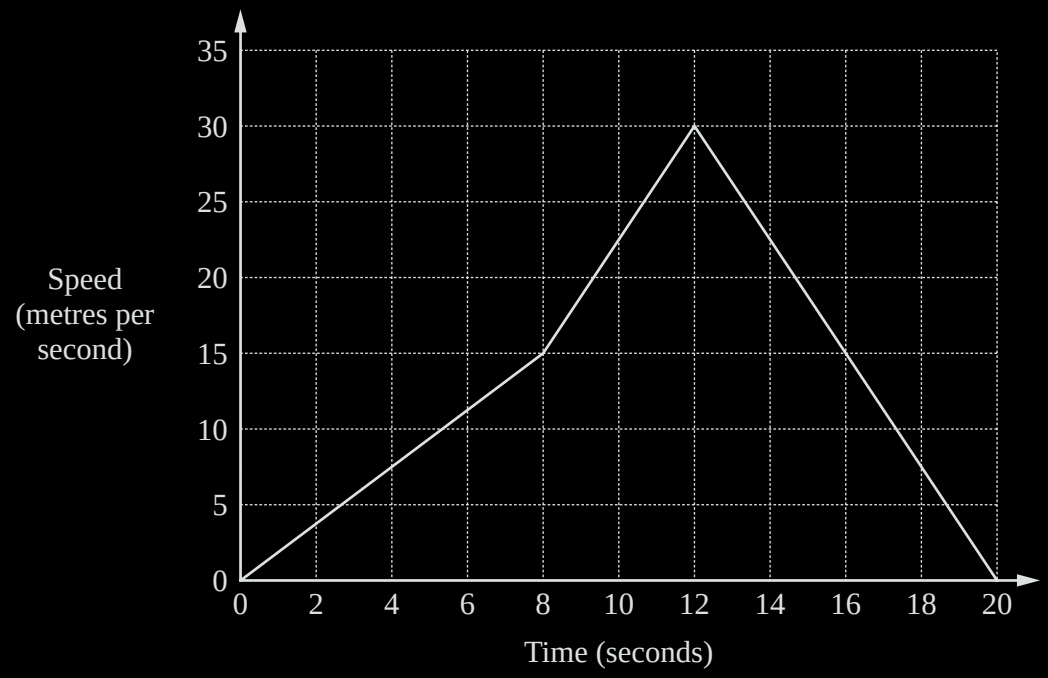
*Answer* $x$ = .....

$y$ = .....[3]

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**13** The diagram is the speed-time graph of the first 20 seconds of a motorcyclist's journey.




**(a)** Calculate the motorcyclist's retardation during the final 8 seconds.

**(b)** Calculate the distance travelled in the 20 seconds.

*Answer* (*a*) ..... $m/s^2$ [1]

(*b*) ..... m [2]

**14** **(a)** A jar contained 370 g of jam.
Usman ate 30% of the jam.

What mass of jam remained in the jar?

**(b)** In 2006 the population of a town was 30 000.
This was 5000 more than the population in 1999.

Calculate the percentage increase in population.

*Answer* (*a*) ..... g [1]

(*b*) ..... % [2]

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**15** Express as a single fraction in its simplest form

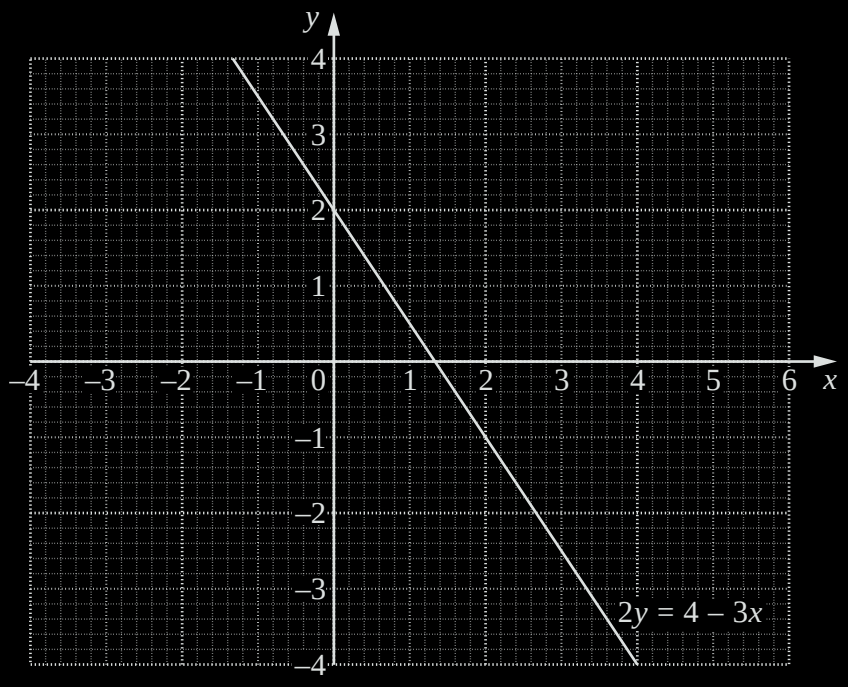
$$\frac{3}{2t-1} - \frac{2}{t+2}.$$

*Answer* .....[3]

---

**16** The diagram below shows the line $2y = 4 - 3x$.

*Answer* (*a*), (*b*)




On this diagram,

**(a)** draw the line $y = \frac{1}{2}x - 2$, [1]

**(b)** shade and label the region, R, defined by the following inequalities.

$$x \geqslant 0 \qquad 2y \leqslant 4 - 3x \qquad y \geqslant \tfrac{1}{2}x - 2$$

[2]

**17** A straight line passes through the points $P$ (1, 2) and $Q$ (5, –14).

Find

**(a)** the coordinates of the midpoint of $PQ$,

**(b)** the gradient of $PQ$,

**(c)** the equation of $PQ$.

*Answer* (*a*) (..... , .....) [1]

(*b*) .....[1]

(*c*) .....[2]

---

**18** The Earth is $1.5 \times 10^8$ kilometres from the Sun.

**(a)** Mercury is $5.81 \times 10^7$ kilometres from the Sun.

How much nearer is the Sun to Mercury than to the Earth?
Give your answer in standard form.

**(b)** A terametre is $10^{12}$ metres.

Find the distance of the Earth from the Sun in terametres.

*Answer* (*a*) ..... km [2]

(*b*) ..... terametres [2]

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**19** **(a)** Factorise completely

**(i)** $15x^2 + 10x$,

**(ii)** $t^2 - 2t - 15$.

**(b)** Solve $4(x - 0.3) = 3(x - 0.2)$.

*Answer* (*a*)(i) .....[1]

(ii) .....[1]

(*b*) $x =$ .....[2]

---

**20** It is given that

$$\mathbf{A} = \begin{pmatrix} 5 & -1 \\ 2 & 3 \end{pmatrix} \qquad \mathbf{B} = \begin{pmatrix} 2 & -2 \\ 0 & 1 \end{pmatrix} \qquad \mathbf{C} = \begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix}.$$

Find

**(a)** $\mathbf{A} - 2\mathbf{B}$,

**(b)** $\mathbf{C}^{-1}$.

*Answer* (*a*) $\begin{pmatrix} & \\ & \end{pmatrix}$ [2]

(*b*) $\begin{pmatrix} & \\ & \end{pmatrix}$ [2]

---

**21** **(a)** Solve $8 - 3t > 14 + t$.

**(b)** Evaluate $x^2 - 6xy + 2y^2$ when $x = 2$ and $y = -3$.

*Answer* (*a*) $t$ .....[2]

(*b*) .....[2]

---

**22** **(a)** The $n$th term of a sequence is $7 - 2n$.

Write down the 23rd term in this sequence.

*Answer* (*a*) .....[1]

**(b)** **(i)** The first five terms of another sequence are

4 7 10 13 16.

Write down an expression, in terms of $n$, for the $n$th term of this sequence.

**(ii)** The first five terms of another sequence are

$$\frac{4}{1} \quad \frac{7}{4} \quad \frac{10}{9} \quad \frac{13}{16} \quad \frac{16}{25}.$$

**(a)** Write down the next term in this sequence.

**(b)** Write down an expression, in terms of $n$, for the $n$th term of this sequence.

*Answer* (*b*) (i) .....[1]

(*b*)(ii)(*a*) .....[1]

(*b*)(ii)(*b*) .....[1]

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**23** A map is drawn to a scale of 1 cm to 3 km.
The diagram below shows the positions of two villages *A* and *B* on the map.

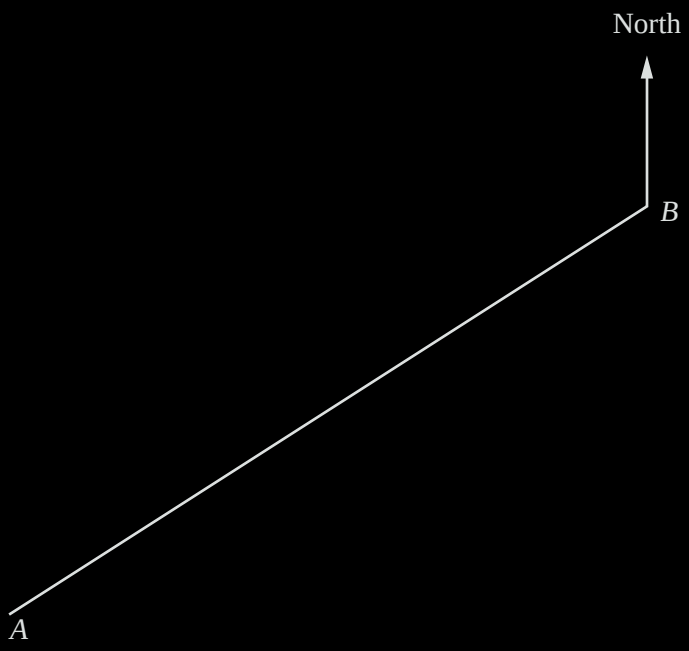
**(a)** **(i)** Write the scale in the form 1 : *n* .

**(ii)** Find the actual distance, in kilometres, between the villages *A* and *B*.

*Answer* (*a*)(i) 1 : .....[1]

(ii) ..... km [1]

*Answer* (*b*), (*c*), (*d*)




[4]

**(b)** A third village, *C*, lies north of the line *AB*.
It is 21 km from *A* and 18 km from *B*.

Using ruler and compasses only, construct triangle *ABC*.

**(c)** Construct the perpendicular bisector of *AB*.

**(d)** A petrol station is to be built so that it is equidistant from *A* and *B* and 9 km from *C*.

Mark with letters *F* and *G* the two possible positions of the petrol station.

**24** **(a)** Fifty students were asked how many books they each took to school on Monday. The results are summarised in the table below.

| Number of books | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|---|
| Frequency | 10 | 11 | 8 | 3 | 6 | 7 | 4 | 1 |

**(i)** Write down the median.

**(ii)** Calculate the mean number of books.

**(iii)** What is the probability that two students, chosen at random, both took 5 books to school?
Give your answer as a fraction in its simplest form.

*Answer* (*a*)(i) .....[1]

(ii) .....[3]

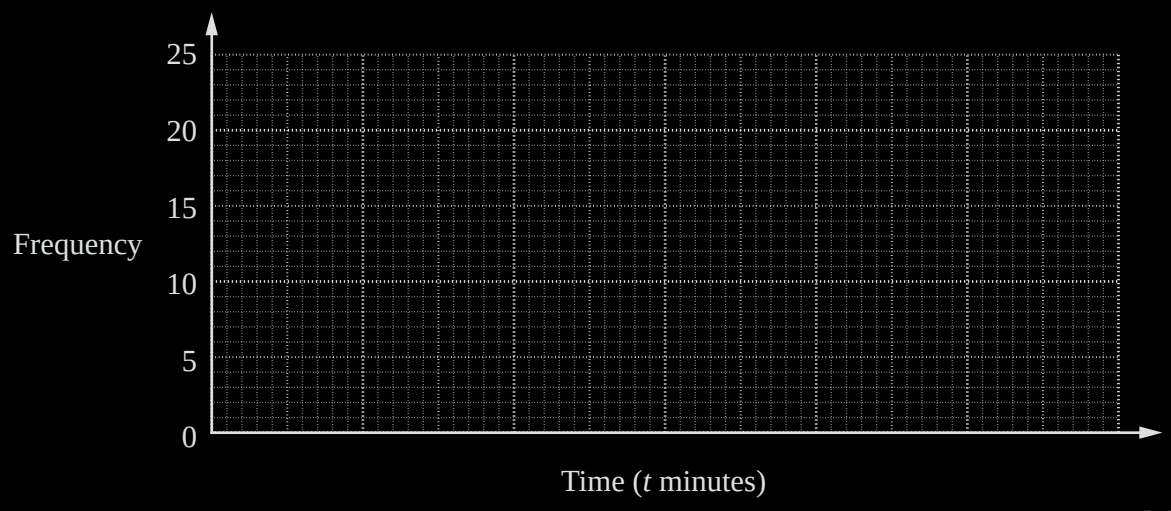
(iii) .....[2]

**(b)** The fifty students were also asked how long they each took to travel to school. The results are summarised in the table below.

| Time of travel ($t$ minutes) | $4 \leqslant t < 6$ | $6 \leqslant t < 8$ | $8 \leqslant t < 10$ | $10 \leqslant t < 12$ |
|---|---|---|---|---|
| Frequency | 21 | 11 | 13 | 5 |

Draw a frequency polygon on the grid below to illustrate this data.

*Answer* (*b*)




[2]

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